# Estimation of Monge Transport Maps with Empirical Brenier Potentials

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## Abstract

We analyze a statistical estimator for Monge transport maps: solutions to the quadratic optimal transport problem. For absolutely continuous source measures, this map is uniquely defined as the gradient of a convex function, a result known as Brenier theorem. Without absolute continuity, the problem is relaxed, maps are replaced by coupling measures, and optimal couplings are supported on the subdifferential of a convex function: a Brenier potential. This characterization is the basis for our Monge transport map estimator, for measures known only through finite samples. The resulting Brenier potential has a simple closed form expression based on the dual solution of the discrete sampled problem. We exhibit convergence rates for this estimator based on a new error bound for the dual quadratic optimal transport problem. Our methodology does not rely on smoothness or continuity of the Monge transport map and requires no computation beyond primal-dual solutions of the discrete, finite dimensional, linear program transport problem.

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